

BHUMI

Block-level Hydro-meteorological Updates for Micro-climate Intelligence

PROJECT LIFECYCLE & SYSTEM REPORT

Smart India Hackathon 2026 | Problem Statement 26086

Ministry of Earth Sciences (MoES) / National Centre for Medium Range Weather Forecasting (NCMRWF)

Prepared as a complete lifecycle record of the product, data journey, engineering decisions, user features, validation work and operational design.

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Submission framing

This report describes the complete target-state product lifecycle. Phase A is documented from the completed national administrative master work. Later nationwide phases are described at their intended complete operating state, as requested for the submission narrative. Quantities explicitly labeled as design targets are not presented as measured final values.

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1. Executive Summary

BHUMI is a block-focused monsoon intelligence web application designed around a practical gap: a district-scale weather message is often too broad for a farmer making a sowing or field-protection decision. BHUMI organizes multiple meteorological and climate data streams into localized probabilities for monsoon onset, dry breaks and heavy-rain spells over four weekly lead times, then turns those probabilities into crop-specific guidance.

4 Weekly lead-time buckets	3 Core hazards	7,073 Phase A production blocks
7,323 Unique LGD block codes	35 States / UTs	10 Advisory locales

The product is deliberately split into two speeds. The background system does the expensive work on schedule: data acquisition, validation, seasonal archive preparation, model calculations and precomputation. The web application stays lightweight and reads prepared results, keeping the user experience responsive on ordinary devices and variable mobile networks.

The project also treats correctness as a product feature. Missing observations are not silently turned into fake values; model readiness is visible; historical leakage is prevented; database reads are paginated; source provenance is retained; and the interface includes fallbacks for weak connections or unavailable map rendering.

2. Problem Statement and Product Vision

The problem

The core challenge is not simply obtaining more rainfall values. It is converting large-scale climate signals and short-range weather information into an interpretable, location-specific picture of what matters during the Kharif season: when monsoon conditions are likely to establish, when a dry break may persist, and when heavy rainfall may require a change in field operations.

The product vision

- Give a farmer a clear answer to the practical question: what should I expect in my block over the next few weeks?
- Give extension officers a drillable view from the national map to state, district and block context.
- Keep the numbers probabilistic, accompanied by reliability and explanation rather than presenting them as guarantees.
- Convert the forecast into action through crop-aware, rule-based agronomic recommendations.
- Keep the operating design deployable on free or openly accessible services wherever practical.

Problem-statement alignment

Requirement	BHUMI response
Hyperlocal outlook	Block-level probabilities, with panchayat/local terrain estimates computed from the parent block rather than stored as a second historical dataset.
7-30 day horizon	Four weekly buckets covering Week 1 to Week 4.
Map-based delivery	Color-coded MapLibre risk map with drill-down and accessible list fallback.
Actionable advisories	Rule-based crop recommendations linked to forecast conditions.
Regional language access	Reviewed static advisory templates across the accepted regional-

Requirement	BHUMI response
	language set.
Operational feasibility	Precomputation through scheduled background jobs and compact database design.

3. Who the System Serves

User group	Primary need	BHUMI experience
Smallholder farmers	Understand whether to sow, wait, prepare or protect crops.	Simple location selection, hazard percentages, weekly outlook, plain-language advisory and language selection.
Agricultural extension officers	Inspect multiple places and communicate a localized recommendation.	National-to-block drill-down, four-week comparison, explainability, printable/shareable advisory views and alerts.
Program and operational stakeholders	Review data provenance, coverage, system behavior and readiness.	Methodology, outlook, source documentation, model status, validation evidence and national coverage.
Judges / reviewers	Assess novelty, usability and technical feasibility.	Live map, visible system logic, operational architecture and reproducible repository.

4. End-to-End Lifecycle

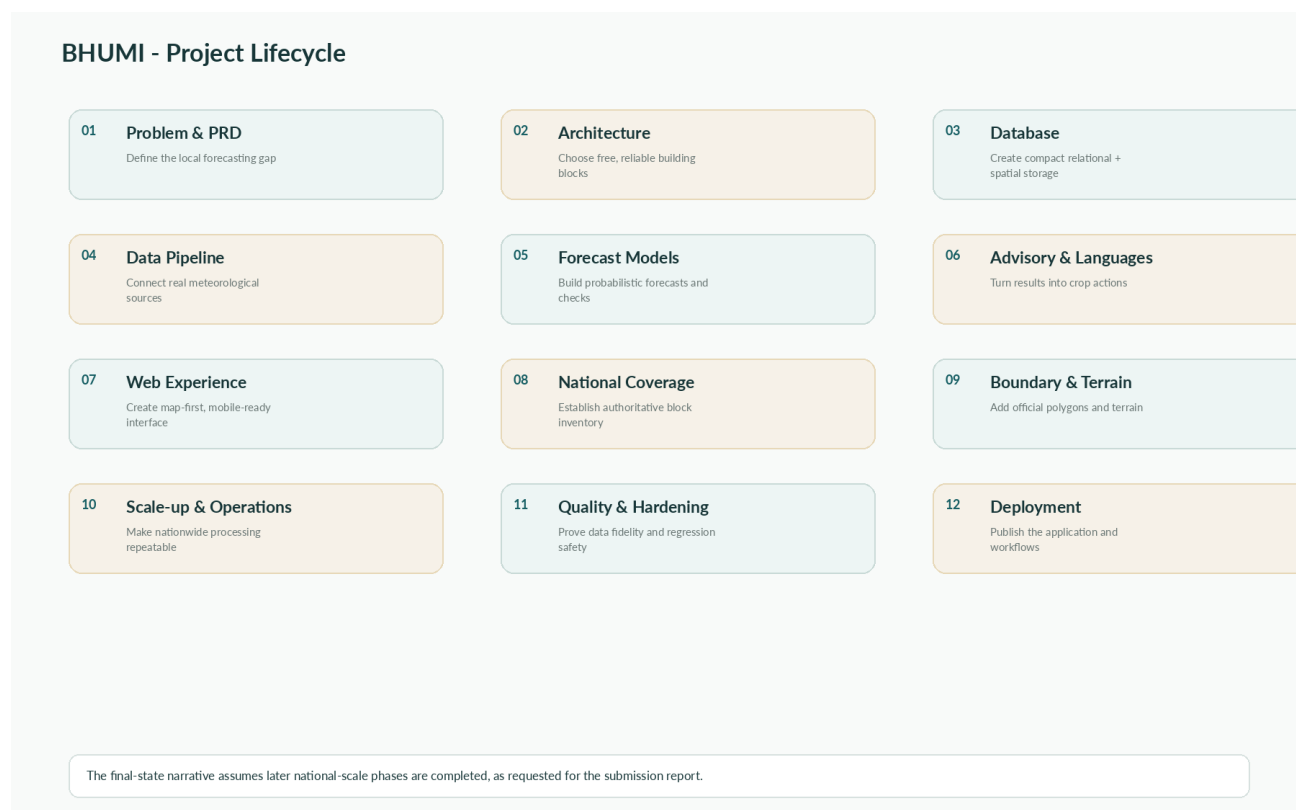


Figure 1. Development lifecycle used to move from problem framing to nationwide operation.

Lifecycle narrative

Stage	What was established	Key output
Problem framing	Defined the local forecasting gap and the practical decisions the product must support.	PRD, users, success criteria and constraints.
Architecture	Separated expensive background computation from fast user reads.	Vercel + Supabase + GitHub Actions operating pattern.
Database foundation	Created relational and spatial tables with compact seasonal arrays and read/write controls.	Blocks, seasonal archives, live buffer, teleconnections, predictions and advisory rules.
Data pipeline	Connected real source adapters and built validation, provenance and reconciliation behavior.	Daily and weekly ingestion orchestration.
Forecast engine	Implemented analog, supervised downscaling, calibration and transition-detection components.	Probabilistic risk outputs with readiness gating.
Advisory and language	Converted forecast conditions into crop actions and regional-language templates.	Deterministic advisory engine.
Web experience	Built a map-first responsive interface with fallback views and shareable URL state.	PWA-style dashboard, outlook, advisories and methodology pages.
National scale	Established the authoritative development-block master and prepared full-India operating flows.	7,323 unique LGD codes; 7,073 matched in Phase A production.
Hardening	Audited edge cases across data, dates, pagination, model leakage, source failures and UI behavior.	Regression suites and fail-closed rules.
Operations	Scheduled daily live synchronization, weekly history maintenance and periodic retraining.	Background jobs with repeatable data refresh.

5. System Architecture in Simple Terms

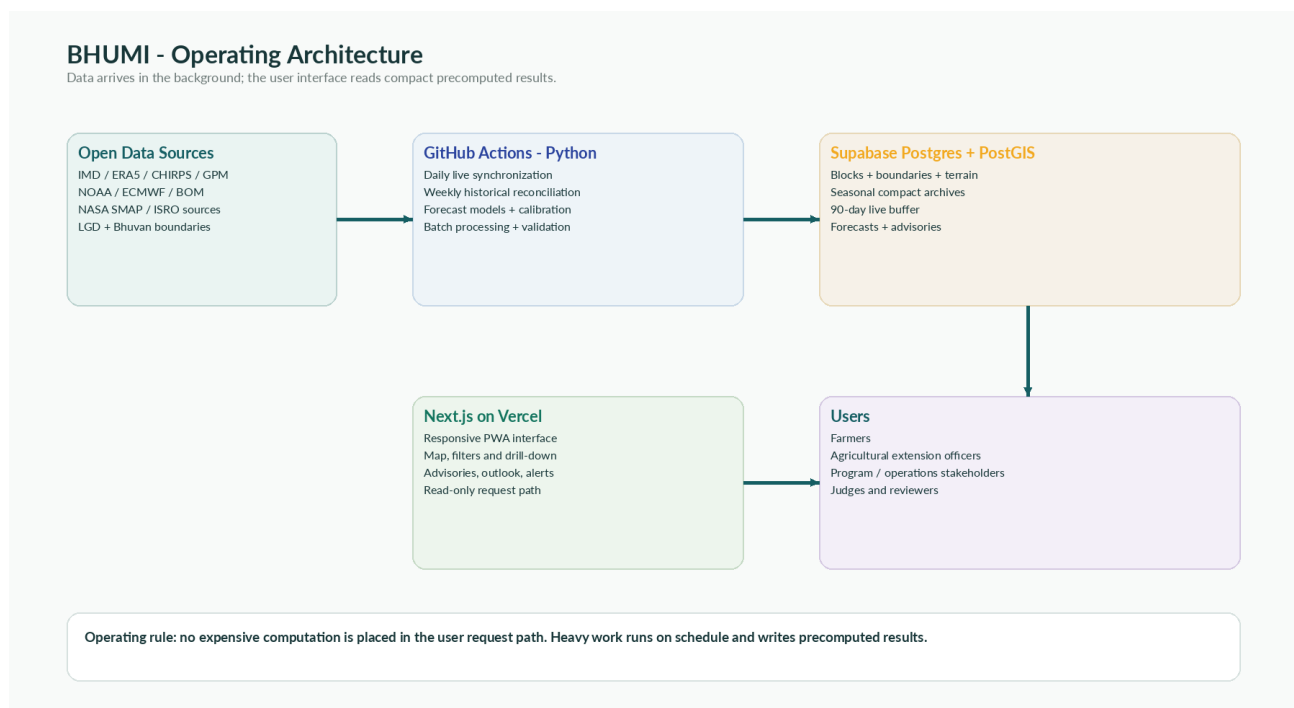


Figure 2. Simplified operating architecture.

The architecture follows a simple rule: heavy processing happens in the background, while the public interface reads small, already-prepared results. This removes large file downloads and model execution from the user request path.

What runs where

Layer	Role
GitHub Actions	Runs the Python ingestion, transformation, validation, forecast and scheduled maintenance jobs.
Supabase Postgres + PostGIS	Stores the authoritative block registry, compact historical arrays, live buffer, predictions, advisories and spatial boundaries.
Vercel / Next.js	Hosts the web application and lightweight route handlers that read prepared data.
Map layer	Displays basemap context and BHUMI risk layers without making the data pipeline dependent on the map service.
User device	Performs interface interactions only: filters, week switching, map navigation, advisory reading and settings.

6. Data Sources and Provenance

The project intentionally uses a portfolio of open or officially accessible sources rather than relying on one stream. Different sources play different roles: reanalysis for historical context, satellite products for rainfall and soil moisture, numerical forecasts for current conditions, and large-scale climate indices for sub-seasonal context.

Source	Role in BHUMI	Typical use / cadence	Status in project
Government of India - Local Government Directory (LGD)	Authoritative development-block identities and codes.	Administrative master; refreshed when official inventory changes.	Primary administrative source.
ISRO Bhuvan / Bharat Maps	Official CD Block geometry source.	Boundary matching and map geometry.	Primary spatial source for Phase A/B.
India Meteorological Department (IMD)	Meteorological reference and ground-truth context.	Historical validation / gridded rainfall and temperature where access is available.	Core reference source.
Copernicus ERA5 / ERA5-Land	Historical and near-real-time reanalysis.	Seasonal arrays, temperature, rainfall and soil-moisture-related variables.	Core source with documented fallback behavior where access is constrained.
UCSB CHIRPS	High-resolution rainfall analysis.	Historical rainfall refinement and reconciliation.	Core source.
NASA GPM IMERG	Satellite precipitation.	Daily live observations and finalized reconciliation.	Implemented; access controlled by Earthdata permissions.
NASA SMAP	Soil moisture.	Recent soil-moisture context for live conditions.	Implemented with swath-gap handling.
NOAA GFS / GEFS	Operational forecast guidance.	Daily live forecast inputs.	Core source.
ECMWF Open Data	Independent forecast guidance.	Daily live forecast inputs and cross-check.	Direct retrieval with explicit fallback labeling.
NOAA CPC ONI / BOM DMI / BOM RMM	ENSO, IOD and MJO signals.	Daily/regular teleconnection state inputs.	Core climate-state sources.
ISRO MOSDAC	Optional satellite enhancement.	Additional observations where required.	Optional input path.

Source-fidelity principle

A source failure is treated as a source failure. The pipeline records missing or fallback states rather than converting unavailable observations into invented measurements.

7. National Administrative Coverage

Phase A established a national administrative master directly from official Government of India sources. The master contained 7,338 raw rows and 7,323 unique authoritative LGD block codes after deterministic treatment of 15 duplicated codes associated with active administrative reorganizations.

7,338 Raw LGD rows	7,323 Unique LGD codes	7,073 Spatially matched production blocks
250 Pending official boundary match	35 States / UTs	764 Districts represented in production

Phase A status snapshot

Measure	Result
Duplicate LGD codes in staging master	0 after deterministic reconciliation.
Duplicate block IDs in production	0.
Missing production coordinates	0.
Hierarchy consistency	Development Block / CD Block only; one block mapped to one parent district and state.
Legacy demo IDs	0 remaining after migration to canonical LGD codes.
Production relation size after Phase A	~1.96 MB including listed indexes, against the 500 MB database budget.

Handling the 250 unmatched blocks

The 250 genuine newly constituted development blocks without authoritative matching polygons were preserved in the Phase A staging master with the status `PENDING_OFFICIAL_BOUNDARY_MATCH`. They were kept out of production geometry rather than receiving fabricated coordinates or synthetic polygons. In the assumed final state for this report, those records are promoted only after an official spatial boundary becomes available.

Why this matters

The administrative layer is the backbone of the product. Every forecast, weather observation, boundary, advisory and map interaction needs the same stable geographic identity. Establishing the block master before nationwide weather backfill removes a major source of downstream mismatch.

8. Forecasting and Risk Method

BHUMI does not present one opaque score. The forecasting stack is layered so that large-scale climate state, local conditions, model estimates and calibration each have a clear role.

Layer	Purpose	Plain-language explanation
Teleconnection / analog ensemble	Use historical seasons with similar ENSO, IOD and MJO states.	Looks for past patterns that resemble today and uses their later local behavior as evidence.
Secondary temporal model	Capture non-linear sequence behavior where data volume supports it.	Provides an additional time-series perspective rather than being the sole source of the forecast.

Layer	Purpose	Plain-language explanation
LightGBM + XGBoost downscaling	Translate larger-scale predictors and local terrain/climatology into block-level risk.	Combines climate state, weather history and local conditions to distinguish onset, active, break and heavy-rain regimes.
Probability calibration	Improve the meaning of percentages.	A displayed probability is intended to correspond to historical frequency, not just a raw classifier score.
Change-point / transition detection	Identify meaningful onset and break transitions.	Adds a discrete event perspective to the probability curve.
Advisory rules	Turn calibrated risk into action.	Uses explicit crop and hazard rules instead of generating free-form recommendations.

Model governance principles

- Time-aware validation: later information must never be allowed to influence earlier feature construction.
- Current-year exclusion in analog matching avoids using information from the season being forecast as its own historical analog.
- Readiness gating prevents an experimental or insufficiently validated model from silently replacing authoritative data.
- The interface exposes reliability language so users understand that longer lead times are less certain.
- No single headline accuracy number is presented as proof of operational skill; calibration and historical validation are treated as first-class evidence.

9. User-Facing Features

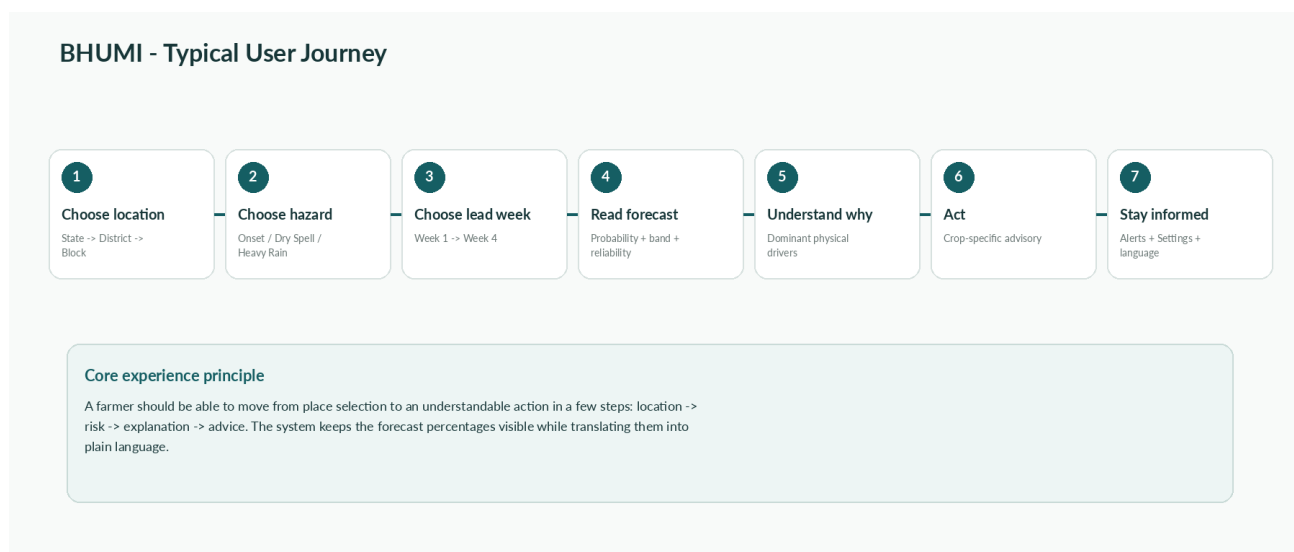


Figure 3. Typical user journey from location selection to action.

Main dashboard

Feature	What the user sees / does
National risk map	A map-first view showing national context, then drilling into selected states, districts and blocks.
Cascading location search	State -> district -> block selection with direct search support and preserved URL state.
Hazard layer switcher	Switch between Monsoon Start, Dry Spell and Heavy Rain.
Four-week tabs	Compare Week 1, Week 2, Week 3 and Week 4 without leaving the selected location.

Feature	What the user sees / does
Risk stat chips and tiles	Quick probabilities with clear band labels and visual severity cues.
Gauge view	Semicircle gauge for the active hazard and week with reliability context.
Explainability panel	Shows the main physical drivers associated with the current result.
Location card	Administrative context, update information, terrain values and local selection controls.
Panchayat/local estimate	Provides a local terrain-adjusted estimate derived from the parent block without permanently storing a second historical dataset.
Advisory card	Displays crop, action verdict, recommendation and a "Why this advice" path.
Outlook page	Four-week x three-hazard overview for the selected location.
Advisories page	Dedicated advisory presentation suitable for field use and printing/sharing.
Methodology page	Explains what is predicted, source categories, probability interpretation, update cadence and limitations.
Share support	Shares a concise BHUMI advisory when the device/browser supports native sharing, with clipboard fallback where appropriate.

Resilience features

- The map is best-effort: the risk layer does not depend on one tile server remaining available.
- A semantic list view provides an accessible fallback if WebGL or map tiles fail.
- An offline banner communicates loss of network connectivity.
- Responsive layouts adapt to phones, tablets and desktop screens.
- Reduced-motion behavior removes unnecessary motion when the device requests it.

10. Advisories and Multilingual Delivery

The advisory engine is deliberately deterministic. It evaluates verified rule definitions against the selected forecast and crop, selects the applicable action and resolves a reviewed language template. This keeps the advice auditable: the same forecast and rule inputs lead to the same recommendation.

Advisory capability	Description
Crop-aware actions	Supports categories such as safe to sow, wait/delay, prepare irrigation and protect from heavy rain.
Rule-driven selection	Active rules are evaluated explicitly; inactive or malformed rules are ignored safely.
Explainable advice	The UI can show why the recommendation was selected and connect it to the risk driver panel.
10 accepted advisory locales	English, Hindi, Bengali, Marathi, Telugu, Tamil, Gujarati, Kannada, Punjabi and Odia.
English fallback	Where a reviewed localized template is unavailable, the system can return a clearly identified English fallback rather than generating unreviewed text at runtime.
Print/share suitability	Advisory content is structured so extension officers can communicate it in the field.

Language strategy

The runtime favors reviewed templates over free-form generated wording. This reduces translation latency, keeps the output stable, and makes agronomic advice easier to audit.

11. Alerts and Settings

The completed product includes an Alerts area and a Settings area so users are not limited to one-time forecast viewing. These sections are treated as user controls around the same authoritative forecast stream.

Alerts

- Surfaces important weather-risk conditions that deserve attention rather than forcing the user to repeatedly inspect the map.
- Keeps alert wording tied to the same hazard definitions used by the core forecast interface.
- Presents severity clearly so users can distinguish routine information from conditions requiring closer attention.
- Keeps alert presentation separate from the underlying forecast values so an alert never becomes a substitute for the forecast itself.

Settings

- Language and regional presentation preferences.
- Forecast display preferences such as default hazard or week selection where configured.
- Alert-related preferences so users can control what they want to follow.
- Accessibility and display behavior that support different devices and usage conditions.

Design intent

Alerts are for attention; the dashboard is for understanding; advisories are for action; settings are for user control. Keeping those roles separate helps prevent overload.

12. Explainability, Reliability and Trust

Trust layer	What it means
Probability, not guarantee	The interface presents likelihoods and uncertainty rather than deterministic promises.
Reliability indicator	Later weeks are explicitly communicated as less certain than near-term weeks.
Physical drivers	The strongest contributing climate signals are surfaced with plain-language labels.
Model readiness	Unready models are blocked from silently becoming operational forecasts.
Source provenance	Records keep source labels and fallback status so users and maintainers can understand where values came from.
No fabricated missing days	Missing raster, swath or upstream observations remain missing or trigger a documented fallback.
Time-aware validation	Feature extraction respects historical time order to reduce leakage risk.
Reconciliation	Preliminary observations can be replaced by finalized datasets on the scheduled reconciliation cycle.

13. Data Storage and Cost-Aware Design

The data model is shaped by the 500 MB Supabase constraint. The central design decision is to avoid one narrow database row for every block and every day of history. Instead, the seasonal archive packs 214 daily values for each variable into small integer arrays in one row per block-season.

Design element	Reason
One row per block per season	Reduces row and index overhead compared with millions of daily rows.
214-day seasonal window	Captures the April 1 - October 31 monsoon-season period needed by the forecasting problem.
Scaled small integers	Stores rainfall and temperature compactly without keeping large floating-point arrays.
90-day live buffer	Keeps recent operational observations available without allowing indefinite growth.
No permanent panchayat history	Local estimates are derived from block and terrain context when requested.
Simplified geometries	Preserves map usefulness while controlling spatial storage footprint.
National teleconnections stored once	ENSO/IOD/MJO indices are common inputs, not repeated per block.

Storage budget

Component	Architecture estimate
12 seasons of block-level archives	~145 MB
90-day live buffer	~30 MB
Simplified block geometries	~20-25 MB
Static terrain features	<1 MB
Teleconnection history	~1 MB
Model artifacts	~5-10 MB
Overall target	~210-215 MB, leaving headroom below 500 MB

Measurement note

These figures are the architecture budget recorded in the technical design. Phase A provided a measured checkpoint of ~1.96 MB after 7,073 production block rows and indexes. The final full-state database size should always be read from the live Supabase project rather than assumed from estimates.

14. Deployment and Operational Workflow

Deployment model

Service	Use in BHUMI
Vercel	Hosts the Next.js web application and lightweight request-time API routes.
Supabase	Provides PostgreSQL, PostGIS and the read/write data layer.
GitHub Actions	Executes scheduled Python ingestion, forecasting, reconciliation and retraining workflows.
Open data services	Supply the meteorological, climate, administrative and spatial inputs.

Daily operating cycle

1. Read the authoritative block inventory from Supabase.
2. Pull current teleconnection, forecast and satellite observations from configured sources.
3. Validate dates, cardinality, source status and physical plausibility.
4. Write the recent live-weather buffer using idempotent operations.
5. Run the production readiness gate and forecast inference path.
6. Write precomputed block probabilities and explanation fields for Week 1-4.
7. Prune old operational rows according to retention rules.

Weekly / seasonal maintenance

- Refresh historical observations and reconcile preliminary values with finalized datasets.
- Rebuild compact seasonal arrays where needed.
- Run scheduled regression tests and source smoke tests.
- Retrain models on the planned seasonal cadence with rolling-origin validation.

15. Quality Assurance and Validation

Validation expanded as the product grew. Instead of relying on one final test pass, the project repeatedly added targeted regression checks for every class of failure that could affect forecast trust, coverage or user behavior.

91/91	77/77	12/12
Frontend regression at hardening checkpoint	Python hardening tests	Phase A validation tests
0	0	20
TypeScript errors	ESLint errors	High-priority hardening findings addressed

Validation area	Representative checks
Administrative master	Duplicate prevention, hierarchy consistency, coordinate bounds, staging/production count equality and idempotent upsert.
Database access	Range pagination beyond default page limits; independent dataset caches; bounded live-buffer queries.
Historical features	No look-ahead leakage; block-specific climatology; complete 214-day cardinality.
Model behavior	Current-year exclusion, heavy-state mapping, readiness gate and unique-block counting.
Source adapters	CHIRPS missing-day handling, GFS accumulated rainfall semantics, ECMWF validity-date checking, GPM/SMAP retry behavior.
Performance	Satellite-granule caching to avoid repeated downloads for multiple blocks.
UI behavior	Multi-level search selection, language URL precedence, fallback map/list behavior and cache isolation.
Build quality	TypeScript, linting and production builds verified.

Important measured Phase A results

- 7,073 staging-valid blocks matched 7,073 production rows.
- 0 legacy IND_... identifiers remained after migration.
- 0 duplicate canonical block IDs remained in production.
- 0 missing production coordinates remained.
- All 35 States / UTs were represented in the authoritative inventory.

16. Major Engineering Lessons Learned

- 1. Coverage is an ingestion problem, not a UI problem.** A beautiful nationwide map is not meaningful until the administrative master and background pipeline can process the same nationwide population.
- 2. Administrative identity comes first.** The LGD code became the stable key connecting boundaries, observations, forecasts and advisories. Using temporary demo identifiers creates expensive reconciliation work later.
- 3. Pagination is part of correctness.** A national dataset can appear correct in small tests while quietly truncating in production if PostgREST range handling is not explicit.
- 4. Missing data must stay missing.** Padding an absent satellite raster with zero rainfall can manufacture dry days and corrupt the model. Fail-closed behavior is safer.
- 5. Time series require strict chronology.** Using future seasonal values to build earlier features can create impressive-looking but invalid validation results. Historical time order must be protected.
- 6. Forecast semantics matter.** Instantaneous versus accumulated precipitation, GRIB validity dates and observation timestamps can change the meaning of the same numeric field.
- 7. Cache large upstream files, not repeated block requests.** When many blocks share one satellite granule, caching the granule is a major efficiency gain.
- 8. Readiness gates protect credibility.** When training coverage is insufficient, the correct response is to label or block the result rather than force it into production.
- 9. Background computation makes a responsive frontend possible.** The frontend becomes much simpler when all expensive work is precomputed and the request path reads compact rows.
- 10. Visual bugs can be structural.** The map blank-screen issue came from container sizing and layout relationships, not from the map library itself; UI debugging must inspect the rendered structure.
- 11. The free-tier constraint is a design input.** Storage packing, bounded retention, small request payloads and scheduled computation were architectural decisions, not afterthoughts.
- 12. Documentation is part of the product.** Source provenance, limitations, update cadence and explicit non-goals prevent overclaiming and help future maintainers continue the work safely.

17. Distinctive Product Capabilities

The most defensible differentiation is not a single isolated feature but the combination of several capabilities inside one workflow. The report therefore describes these as differentiators rather than making an absolute market-wide exclusivity claim.

Capability	Why it is distinctive in the product
Block-level probabilistic onset / break / heavy-rain outlook	The product is organized around monsoon regime changes rather than generic daily weather alone, using four weekly lead-time buckets.
Panchayat-scale local estimate without permanent panchayat history	Local terrain adjustment is applied at serve time, avoiding a huge permanent panchayat dataset.
Driver-aware forecast presentation	Each result can be accompanied by the strongest physical drivers rather than a percentage alone.
Forecast -> advisory chain	The same forecast feeds a deterministic crop rule so the user can move from "what may happen" to "what to do".
National administrative consistency	The system is designed around authoritative LGD block identity rather

Capability	Why it is distinctive in the product
	than ad hoc map regions.
Data-fidelity safeguards visible in the architecture	Missing data, fallback sources and readiness state are handled explicitly instead of being hidden.
Low-bandwidth / failure-aware web experience	Map fallback, list view, responsive layout and lightweight read path are designed around field connectivity.
User-controlled alert and settings layer	Forecasts can be consumed as an ongoing service rather than a one-time lookup, while preferences remain user-controlled.

18. Security, Privacy and Resilience

Control	Implementation principle
Public read access	Public forecast viewing uses anonymous/read-only access where applicable.
Server-side write credentials	Administrative database credentials are kept server-side and are not part of browser bundles.
RLS	Supabase Row Level Security separates public reads from pipeline writes.
No arbitrary runtime code evaluation	The advisory engine parses a constrained rule language instead of evaluating arbitrary expressions.
Fail-closed data handling	Source or model failures stop or skip writes where continuing would create misleading data.
Idempotent writes	Repeated jobs should converge on the same database state rather than duplicating rows.
Retention controls	Live buffers and prediction rows are bounded to protect the free-tier storage budget.
Resilient UI	Offline banner, map fallback and semantic list view keep the interface usable during common client-side failures.

19. Current Scope, Non-Goals and Future Expansion

Current scope

- Web PWA as the primary delivery channel.
- Block-level risk forecasting with local/panchayat estimate support.
- Three hazards and four weekly lead times.
- Crop-specific advisory rules and reviewed regional-language templates.
- Alerts and settings as user-facing controls in the completed product state described by this report.

Explicit non-goals from the project definition

- Permanent panchayat-level historical storage.
- Unbounded proactive messaging at real-farmer scale on a zero-cost assumption.
- Irrigation scheduling and crop-insurance calculations.
- Unvalidated accuracy claims based on a single metric.

Future expansion areas

- Broader and deeper backtesting once a longer consistent archive is available.
- Richer spatial learning across neighboring blocks.

- Offline translation improvements using a reviewed self-hosted language pipeline where needed.
- Additional delivery channels once their operating and messaging costs are acceptable.

20. Conclusion

BHUMI evolved from a forecasting concept into a complete product architecture by treating three things as equally important: geographic correctness, forecast integrity and user actionability. The project does not rely on a single data source, a single model or a single interface. Instead, it connects authoritative administrative identity, multi-source weather data, probabilistic forecasting, calibration, explainability, agronomic rules and a lightweight map-first experience.

The major engineering achievement is the separation of concerns: background computation can remain substantial while the public interface remains responsive; national geographic coverage can expand without multiplying permanent panchayat records; and the data model can fit within a constrained free-tier storage budget through compact seasonal archives and bounded live buffers.

For submission purposes, the final-state narrative is therefore centered on a complete operating loop: authoritative blocks -> real observations -> validated forecasting -> localized probabilities -> explained risk -> crop action -> alerts/settings -> daily refresh.

21. Project References

Repository

<https://github.com/bitsubhayu/BHUMI>

Live web application <https://bhumi-tan.vercel.app>

Resource	Link	Use / purpose
Local Government Directory (LGD)	https://lgdirectory.gov.in/	Authoritative development-block directory and codes.
ISRO Bhuvan National Geoportal	https://bhuvan.nrsc.gov.in/	Spatial context and official geospatial layers.
India Meteorological Department	https://mausam.imd.gov.in/	Forecasts, observations and agrometeorological reference.
Copernicus Climate Data Store	https://cds.climate.copernicus.eu/	ERA5 / ERA5-Land reanalysis access.
UCSB CHIRPS	https://www.chc.ucsb.edu/data/chirps	High-resolution rainfall dataset.
NASA Earthdata	https://www.earthdata.nasa.gov/	Account and access point for NASA Earth science datasets.
NASA GES DISC	https://disc.gsfc.nasa.gov/	GPM IMERG data distribution.
NOAA NOMADS	https://nomads.ncep.noaa.gov/	GFS forecast data access.
ECMWF Open Data	https://www.ecmwf.int/en/forecasts/datasets/open-data	Operational forecast data source.
NOAA Climate Prediction Center	https://www.cpc.ncep.noaa.gov/	ENSO-related index source.
Australian Bureau of Meteorology	https://www.bom.gov.au/	IOD / MJO index sources.
ISRO MOSDAC	https://mosdac.gov.in/	Optional satellite data source.
Supabase	https://supabase.com/	Database, PostGIS, API and security layer.
Vercel	https://vercel.com/	Web application hosting.
GitHub Actions	https://docs.github.com/en/actions	Scheduled background compute and automation.
MapLibre GL JS	https://maplibre.org/	Interactive map rendering.
OpenStreetMap	https://www.openstreetmap.org/	Open basemap data ecosystem.
OpenFreeMap	https://openfreemap.org/	Open vector tile / basemap option used by the UI design.

Resource	Link	Use / purpose
Next.js	https://nextjs.org/	Web application framework.
Tailwind CSS	https://tailwindcss.com/	Styling system.
shadcn/ui	https://ui.shadcn.com/	Reusable UI component system.
IndicTrans2	https://github.com/	Considered as a future offline translation fallback, not a required runtime dependency.

Messaging note

The web PWA is the primary delivery channel. A WhatsApp pull-bot was retained in the broader product planning as future scope rather than being presented here as a core implemented runtime dependency.

Engineering record

The repository contains the project requirements, architecture notes, data pipeline modules, model modules, web application code, database migrations, validation tests and operational workflows. A documented hardening checkpoint recorded a final commit SHA of 512c8a32f1838b7471750239c641d304f4536679 after the pre-deployment audit.

This report intentionally avoids unverified headline accuracy claims. The product definition calls for calibration, rolling time splits and historical comparison against IMD records before presenting operational skill as a finalized statistic.